

DRUGS USED IN PARKINSONISM

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OVERVIEW

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- ▶ These are diseases like parkinsonism disease (PD), Huntington disease (HD), multiple sclerosis (MS) and amyotrophic lateral sclerosis
- ▶ These are neurodegenerative in nature characterized by the progressive loss of neuronal function in a particular part of the CNS
- ▶ The signs and symptoms of neurodegenerative disease do not reflect a normal age related loss of brain neuron instead these disease states are a result of an underlying pathologic cause
- ▶ Although the cause of these diseases is not known, evidence suggests the involvement of heredity, autoimmune and environmental factors

PARKINSONISM DISEASE (Paralysis Agitans)

- ▶ Parkinsonism is a disease condition that comes about due to the degeneration of dopaminergic neurons arising from the substantia nigra and projecting to other structures in the basal ganglia

- ▶ It is characterized by the following features;
 1. Resting tremors
 - Involuntary trembling when the limb is at rest
 2. Rigidity
 - Inability to initiate movement

Etiology and pathogenesis of parkinsonism

- ▶ The real cause neurodegenerative diseases largely remain unknown, and mostly heredity appears to have a limited role
- ▶ However, so far identified is a defective gene responsible for autosomal juvenile parkinsonism which usually affects people in their teens and 20s

Oxidative stress theory in Parkinsonism disease

- ▶ This is one of the better known theories on the cause of Parkinsonism
- ▶ In this theory, it is hypothesized that metabolic oxidation of dopamine in the basal ganglia yield highly reactive free radicals that are toxic to dopaminergic neurons and thus leading to their degeneration
- ▶ Free radicals are molecules that lack an electron in the outer orbits and are capable of extracting an electron from other molecules and thereby causing damage
- ▶ It is not understood why some individuals are more susceptible to oxidative stress than others

Etiology of PD cont'd

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- ▶ The basal ganglia is a group of interconnected subcortical nuclei
- ▶ In health individuals, it receives input from the cerebral cortex, process the information and send feedback to the motor area of the cortex leading to coordination of smooth muscle movements
- ▶ In PD patients, the neuronal degeneration disrupts this interplay and affects movement including simple movements like walking
- ▶ Since the basal ganglia also involved in procedural memory and other cognitive functions, patients with PD may have difficulty remembering how to perform learned motor skills like driving
- ▶ The striatum receives input from the cerebral cortex and substantia nigra and sends to the thalamus which then feeds back to the motor cortex through two pathways i.e. the direct also called the excitatory and the indirect inhibitory one
- ▶ In patients with PD, the neurodegeneration leads to the decreased activity in the direct pathway and increased activity in the inhibitory pathway
- ▶ Furthermore, the overall degeneration of dopaminergic neurons leads to relative excess of cholinergic activity in these pathways

Summary of the nature of drugs used in Parkinsonism

- ▶ Parkinsonism is therefore, effectively treated with drugs that can pharmacologically achieve the following;
 - I. Increase dopamine levels
 - II. Inhibit cholinergic activity

DRUGS THAT INCREASE DOPAMINE LEVELS

Levodopa

- ▶ Also called L-dopa or dihydroxyphenylalanine is the precursor of dopamine and is the main treatment of motor dysfunction in PD patients
- ▶ Dopamine itself cannot cross the blood brain barrier and hence this precursor being useful

Mechanism and action

- ▶ In the brain, levodopa is taken up by dopaminergic neurons in the striatum and converted to dopamine by Aromatic amino acid decarboxylase (LAAD)
- ▶ This eventually leads to the increase dopamine released by neurons in PD patients and serving as a form of replacing therapy
- ▶ Levodopa can counteract all signs of PD though the degree and duration of its effectiveness are not optimal in that as the disease progresses, more neurons are lost and the conversion of levodopa to dopamine declines

Indications of levodopa

- ▶ Effective in the management of patients with idiopathic PD
- ▶ Treatment of patients with post encephalitic parkinsonism
- ▶ Treatment of parkinsonism caused by carbon monoxide poisoning, manganese intoxication, cerebral arteriosclerosis

Pharmacokinetics

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- ▶ Levodopa is metabolized by two pathways i.e. L - aromatic amino acid decarboxylase (LAAD) which converts it to dopamine while Catecho-O-methyl transferase (COMT) converts it to 3-O-methyldopa
- ▶ Levodopa is prone to peripheral metabolism by these two enzymes which can lead to reduced concentration reaching the brain tissue
- ▶ Therefore decarboxylase inhibitors like carbidopa and COMT inhibitor like entacapone are separately used in combination with levodopa to help increase the amount of levodopa that reaches the brain tissue
- ▶ Dietary amino acids compete with levodopa thereby, leading to reduced levodopa transport and thus a protein restricted diet is advisable
- ▶ LAAD requires pyridoxine (Vit B6) as a cofactor and thus vit B6 should not be used concomitantly with levodopa as they may enhance peripheral decarboxylation of levodopa

The “Wearing off” and “On Off effect” of levodopa

- ▶ Over time during treatment with levodopa, patients may begin to experience two extremes which are due to reduced concentration of levodopa

Wearing off effect

- ▶ This occurs towards the end of the dosage interval leading to the flaring up of the symptoms

The On Off phenomena

- ▶ This is characterized by severe motor fluctuations that occur randomly

Adverse effects

- ▶ Vomiting especially when used alone
- ▶ Orthostatic hypotension
- ▶ Cardiac arrhythmias which are substantially reduced when combined with carbidopa
- ▶ Psychotic disorders and dyskinesia which can be reduced by lowering the doses of levodopa

Amantadine

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- ▶ This is an antiviral used in the treatment of influenza but also has a beneficial effect in PD
- ▶ It works by increasing the release of nigrostriatal neurons and has also been found to prevent reuptake dopamine by these neurons
- ▶ Amantadine is generally better tolerated than levodopa and dopamine agonist but is less effective

Indication

- ▶ Amantadine is used to treat early or mild case of parkinsonism and as an adjunct for levodopa

Adverse effects

- ▶ Sedation, restlessness, vivid dreams, nausea, dry mouth and hypotension
- ▶ CNS side effects are most likely to occur among the elderly due to their reduced capacity to excrete the drug via the kidneys

Selegiline

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- ▶ This drug is well absorbed from the gut and is partly metabolized to amphetamines
- ▶ This works by inhibiting the MAO-B thereby, preventing the oxidation of dopamine to dihydroxyphenylacetic acid and hydrogen peroxide and thus eventually increasing dopamine in the basal ganglia
- ▶ Selegiline inhibits the progression of PD by inhibiting formation of free radicals or active metabolite of an environmental toxin

Indication

- ▶ Used as a single drug for the treatment of early PD
- ▶ Adjunct to Levodopa-carbidopa for advanced disease
- ▶ Used in the reduction of dose requirement of levodopa and it may improve motor function in patients who experience wearing off and On off difficulties with levodopa

Dopamine agonists (still under drugs that increase dopamine levels)

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Bromocriptine

- ▶ Bromocriptine is a D₂ agonist and D₁ antagonist
- ▶ It serves as an effective adjunct to levodopa in patients who have advanced PD and experience wearing off effects and On and Off motor fluctuations

Adverse effects

- ▶ Presents similar side effects like levodopa
- ▶ Dose related CNS effects of the receptor agonists include confusion, dyskinesia, sedation, vivid dreams and hallucinations

Pramipexole and ropinirole

- ▶ Both act as selective D₂ –receptor agonists
- ▶ In addition, pramipexole act on D₃ receptors and this could be the reason why it is effective in the management of PD
- ▶ Clinical studies have shown that pramipexole and ropinirole can delay the need for levodopa
- ▶ In advanced PD, the newer agonists can reduce the on off period and decrease the levodopa dosage requirement

Acetylcholine Receptor Antagonists

- ▶ Several centrally acting anticholinergic drugs are used in the management of PD
- ▶ Examples are benztropine and trihexyphenidyl hydrochloride which are antagonists at cholinergic muscarinic receptors and also exhibit some antihistamine activity
- ▶ The anticholinergic drugs are generally less effective than dopaminergic drugs but may be helpful as adjunct therapy in combination with levodopa and other dopamine augmenting drugs
- ▶ They are more effective in reducing tremor than in reducing other manifestation of PD though they may reduce the bradykinesias and rigidity

Treatment considerations

- ▶ Optimal management requires that the disabilities of each patient be carefully assessed before drug is recommended
- ▶ Early disease of mild intensity can be managed with exercise, nutrition and education
- ▶ Anticholinergic drugs and amantadine may be helpful in patients whose clinical manifestations are limited to mild tremor and slowness
- ▶ Patients with more severe functional disabilities will benefit from dopaminergic drugs of which levodopa-carbidopa is usually prescribed

END